Qi Liu (Leo)

Contact Information

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Education

Tsinghua University (THU) — September 2018 - June 2021
M.Eng. in Control Engineering — Beijing, China
GPA: 3.9/4.0, Ranking: 6/137
Advisor: Prof. Min Liu
Master Thesis: Research on Methods for Identifying the Tumor-specific DNA Methylation Sites in Peripheral Blood
Relevant Courses: Big Data Systems (A+), Data Mining: Theory and Algorithms (A), Experiment Design and Data Processing (A), Pattern Recognition (A-), Big Data Analytics (A-)

China University of Mining and Technology (CUMT) — September 2014 - June 2018
B.Eng. in Electronic Information Science and Technology — Xuzhou, China
GPA: 3.5/4.0, Ranking: 17/116
Advisor: Prof. Bing Liu
Undergraduate thesis: Research and Implementation of Image Captioning Based on LSTM
Related courses: Software Engineering (A+), Electronic Design (A+), Software Course Design (A+), Computer Organization Principles Experiment (A+), Microcontroller Application Technology (A), Microcomputer Principles and Interfaces (A-), Computer Networks (A-)

Honors and Awards

The Excellent Enterprise Practice Award of the Department of Automation of THU	2020
Tsinghua Comprehensive Excellence Scholarship (Top 5%)	2019
CUMT Excellent Undergraduate Thesis (Top 1%)	2018
CUMT First Prize Scholarship and Outstanding Student Award (Top 5%)	2017
First Prize in The 13th National College Students' Embedded System Design Competition	2017
Second Prize of East China Division in the 3rd National College Students' "Internet+" Innovation and Entrepreneurship Competition	2017

Publications

In Silico drug repurposing pipeline using deep learning and structure based approaches in epilepsy
Lv, X., Wang, J., Yuan, Y., Pan, L., **Liu, Q.**, Guo, J. (2024). In Silico drug repurposing pipeline using deep learning and structure based approaches in epilepsy. Scientific Reports, 14(1), 16562. (IF 3.8)
Zhou, X., Cheng, Z., Dong, M., **Liu, Q.**, Yang, W., Liu, M., Tian, J., & Cheng, W. (2022). Tumor fractions deciphered from circulating cell-free DNA methylation for cancer early diagnosis. Nature Communications, 13(1), 7694. (IF 17.69)

Working Experience

Global Health Drug Discovery Institute (GHDDI) — June 2021 - present
AI Research, Associate Scientist — Beijing, China
Collaborated with members of the Data Science group to develop new algorithms for solving specific problems in drug discovery and to build related computational and software platforms. Representative work:

- "PertGPT" To predict the impact of drug perturbations on single-cell gene expression profiles using **large language models (LLMs)**. Collected 33 million single-cell gene expression altas for pretraining a **scGPT (Single-cell Generative Pre-trained Transformer)** model with 51 million parameters. Fine-tune it for various downstream tasks including drug perturbation prediction.
- "MolGen" Used generative models to create molecules that satisfy specific properties and taking into account the relevant interactions between molecules and proteins. Integrated the graph structure information of molecules (atoms, chemical bonds) with molecule-protein interaction information into a unified fingerprint, and used the **cVAE (Conditional Variational Autoencoder)** model to generate novel molecules that satisfied specific properties.

- "ChemTreeMap" Represent molecular libraries as tree structures to explore the diversity of the library and to compare molecular similarities. Used RDKit to compute 4+ **molecular fingerprints**, generated distance matrix and then built Tree structure using neighbor joining algorithm. Visulized the tree and biological/chemical data using D3.js, and developed the user-friendly API with **FastAPI**.
- "CryoFibrinogen" Enhanced data preprocessing and augmentation techniques were utilized, alongside the combined use of CryoSPARC and Relion, to extract a higher number of potential fibrinogen particles from cryo-EM images. Subsequently, the VAE-based generative model, **CryoDRGN (Cryo Deep Reconstructing Generative Networks)**, was employed to construct the 3D density map of fibrinogen.

Longmaster Information & Tech Co.,Ltd. September 2019 - June 2020

R&D Intern Guiyang, China

Developed an intelligent segmentation system for annotating chest **CT (Computed Tomography) MRI (Magnetic Resonance Imaging)** images. Use **UNet with VGG11** as the backbone, the Jarccard index got 0.91 and 0.92 on the lung segmentation and liver segmentation tasks respectively. Integrated unsupervised models such as KMeans, as well as supervised segmentation models such as FCN (Fully Convolutional Network) and FPN (Feature Pyramid Network). Utilized **Dash and Plotly** for data and segmentation result visualization. Improved the support for **DICOM and NIFTI** in the open-source annotation software LabelMe based on PyQt.

Research Experience

Tsinghua University June 2020 - June 2021

Master's Researcher Beijing, China

Integrated **DNA methylation** data of multiple types of tumors and blood from some tumor-related databases such as **TCGA (The Cancer Genome Atlas), UCSC Xena, GEO (Gene Expression Omnibus)**. Constructed a large-scale DNA methylation dataset containing 14 cancer types. Proposed a method based on category-specific filtering for identifying DNA methylation sites. And a measurement method based on statistical significance and mutual information for measuring DNA methylation sites. Supervised classification models were used to evaluate the performance of the prediction of tumor tissue-of-origin. The accuracy reached to 0.88 for classifying 14 cancer types.

China University of Mining and Technology November 2017 - June 2018

Undergraduate Researcher Xuzhou, China

To let the computer automatically generate natural language descriptions corresponding to the content of an image. Built end-to-end image captioning models by combining **RNN (Recurrent Neural Network)** especially **LSTM (Long Short-Term Memory)** with several **deep convolutional neural networks (CNNs)**. Compared the natural language description performance of the models on various datasets, and developed an application software. Results on the Flisk8k dataset shows that the BLEU-1 (Bilingual Evaluation Understudy) of the one model reached 0.508, and the BLEU-2 reached 0.300.

Leadership Experience

IOT (Internet of Things) & Big Data Institute of CUMT December 2016 - June 2017

Deputy head Xuzhou, China

Assisted the professor in designing laboratory management regulations, managed a student team of more than 20 people, and organized 10+ scientific and technological innovation competitions, lectures, and other events.

Skills

Technical Skills: Proficient with Python, Shell, Git/GitHub; Familiar with PyTorch, R, SQL, C/C++, OpenCV, Web Development and LaTeX

Certifications & Training: IELTS-Test 6.5/9.0 (Listing 6, Reading 7.5 Writing 6 Speaking 6), Certified CAD Engineer, Driving License (C1), Mayo Clinic Healthcare Summit

Languages: Mandarin (native), English (B2 Certified), German (beginner)